

Definition and Scope of Building Technology

Introduction to Building Technology:

- Building technology refers to the application of scientific and engineering principles in the construction of buildings.
- It involves planning, design, construction materials, methods, and maintenance of buildings.
- It helps in creating safe, durable, comfortable, and economical structures.
- Building technology plays a vital role in modern construction.
- Continuous innovation in building technology is essential for future development.

Definition of Building Technology:

- “Building Technology is the branch of engineering and architecture that deals with the technical aspects of building construction, including materials, methods, structural systems, and services used in buildings.”

Objectives of Building Technology:

- To ensure safety and stability of structures
- To improve durability of buildings
- To provide comfort and functionality
- To reduce construction cost and time
- To promote sustainable and eco-friendly construction

Scope of Building Technology:

□ Building technology covers:

- Building planning and design
- Construction materials
- Construction techniques
- Structural systems
- Building services
- Maintenance and repair
- Green and smart buildings

Building Materials:

☐Common Materials Used

- Cement
- Sand
- Bricks
- Steel
- Timber
- Concrete
- Glass
- Stone

Importance:

- Strength and durability
- Cost effectiveness
- Fire resistance
- Thermal insulation
- Easily available

Construction Techniques:

❑ Traditional Techniques

- Masonry construction
- Timber construction

❑ Modern Techniques

- Prefabrication
- Modular construction
- RCC frame construction
- Steel frame construction

Structural Systems:

□Types of Structural Systems

- Load bearing structure
- Framed structure
- Shell structure
- Truss structure

Building Services:

☐Major Services

- Water supply system
- Drainage and sanitation
- Electrical system
- Heating, Ventilation and Air Conditioning (HVAC) system
- Fire protection system
- Communication system

Sustainable Building Technology:

☐Features

- Energy efficient buildings
- Use of renewable energy
- Rainwater harvesting
- Waste management
- Eco-friendly materials

☐Benefits

- Environmental protection
- Reduced energy consumption
- Improved indoor comfort

Advantages of Building Technology:

- Faster construction
- Better quality control
- Improved safety
- Cost reduction
- Increased durability
- Modern appearance and comfort

Challenges in Building Technology:

- High initial investment
- Skilled manpower requirement
- Rapid technological changes
- Environmental concerns
- Maintenance issues

History of Building Technology and Nepalese Building

Introduction:

- Building technology has developed continuously throughout history.
- Building technology has evolved from primitive shelters to modern smart buildings.
- Nepalese buildings reflect the country's culture, geography, climate, and traditions.
- The history of building technology shows the development of construction materials, techniques, and architectural styles.
- Combining traditional knowledge with modern technology can create safe and sustainable buildings.

Early History:

❑ Primitive Period

- Humans used caves, mud, stones, leaves, and wood for shelter.
- Construction was simple and based on locally available materials.

❑ Ancient Civilization

- Egyptians built pyramids.
- Romans developed arches.
- Greeks introduced classical architecture.

Medieval and Industrial Period:

❑Medieval Period

- Development of temples and churches.
- Use of stone masonry and timber framing.

❑Industrial Revolution

- Introduction of steel, cement, and machinery.
- Rise of high-rise buildings and bridges.
- Faster and stronger construction methods.

Modern Building Technology:

❑ Features

- Reinforced Cement Concrete (RCC)
- Steel structures
- Prefabrication
- Smart buildings
- Green building technology

❑ Importance

- Improved safety and durability
- Energy efficiency
- Earthquake-resistant structures

History of Nepalese Buildings:

❑ Ancient Nepalese Architecture

- Influenced by Hindu and Buddhist culture.
- Use of brick, stone, timber, and mud.
- Development of temples, palaces.

❑ Traditional Building Style

- Pagoda style
- Shikhara style
- Stupa architecture

❖ **Pagoda Style Architecture:**

❑ **Characteristics**

- Multi-tiered roofs
- Wide eaves
- Wooden carvings
- Brick masonry walls

❑ **Examples**

- Nyatapola Temple
- Taleju Temple



❖ **Stupa Style**

- Dome-shaped structures
- Buddhist religious monuments
- Symbol of peace and spirituality

❖ **Shikhara Style**

- Tall and curvilinear towers
- Mainly used in Hindu temples

➤ **Examples**

- Swayambhunath Stupa
- Krishna Mandir

Stupa Style



Shikhara Style



Traditional Nepalese Building Materials:

- Stone
- Brick
- Mud mortar
- Timber
- Slate roofing

Advantages:

- Locally available
- Environment friendly
- Good thermal performance

Modern Nepalese Buildings

□ Features

- RCC framed structures
- Concrete and steel construction
- Earthquake-resistant design
- Modern residential and commercial buildings

Causes of Modernization

- Urbanization
- Technological advancement
- Population growth

Earthquake Resistant Construction in Nepal

❑ Importance

- Nepal lies in an earthquake-prone region.
- Safe construction techniques are necessary.

❑ Techniques

- RCC frame structures
- Reinforcement bands
- Proper foundation design
- Quality materials and workmanship

Advantages of Nepalese Traditional Buildings:

- Cultural identity and heritage
- Natural ventilation and lighting
- Eco-friendly construction
- Use of local materials and labor

Challenges in Nepalese Building Construction:

- Earthquake vulnerability
- Rapid urbanization
- High construction costs
- Loss of traditional architecture
- Skilled manpower shortage

Classification of Building Based on Occupancy, Storey and Height

Introduction:

- Buildings are classified according to their use, number of storeys, and height.
- Classification helps in planning, design, safety, and construction management.
- Different building types require different structural and safety provisions.

❑ **Classification Based on Occupancy:**

- Residential buildings (Houses, Apartments)
- Educational buildings (Schools)
- Institutional buildings (Hospitals)
- Assembly Buildings (Auditoriums)
- Business Buildings (Banks)
- Mercantile buildings (Shopping malls)
- Industrial buildings (Factories)
- Storage buildings (Cold storage)
- Hazardous buildings (Gas plants)

❑ **Classification Based on Number of Storeys:**

- Single-storey building (Only one floor above ground)
- Double-storey building (Two floors above ground)
- Multi-storey building (More than two floors)
- High-rise building (Very tall buildings with many floors)

❑ **Classification Based on Height:**

- Low-rise buildings (Up to 10 m)
- Mid-rise buildings (10 m to 25 m)
- High-rise buildings (Above 25 m)

Importance of Building Classification:

- Helps in building design and planning
- Ensures public safety
- Assists in fire protection planning
- Determines structural requirements
- Helps in following building codes and regulations

Thank You